Data dictionary

Source fields

Field	Meaning	Allowed values / type	Role
customerID	Customer identifier; key only, never a feature	unique, non-empty text	key
gender	Recorded gender; audit only, not a model input	Male; Female	audit only
SeniorCitizen	Senior-citizen flag; audit only, not a model input	0; 1	audit only
Partner	Has a partner	Yes; No	feature
Dependents	Has dependents	Yes; No	feature
tenure	Months with the company at the snapshot	non-negative number	feature
PhoneService	Has phone service	Yes; No	feature
MultipleLines	Has multiple phone lines	Yes; No; No phone service	feature
InternetService	Internet service type	DSL; Fiber optic; No	feature
OnlineSecurity	Online security add-on	Yes; No; No internet service	feature
OnlineBackup	Online backup add-on	Yes; No; No internet service	feature
DeviceProtection	Device protection add-on	Yes; No; No internet service	feature
TechSupport	Technical support add-on	Yes; No; No internet service	feature
StreamingTV	Streaming TV add-on	Yes; No; No internet service	feature
StreamingMovies	Streaming movies add-on	Yes; No; No internet service	feature
Contract	Contract term	Month-to-month; One year; Two year	feature
PaperlessBilling	Paperless billing	Yes; No	feature
PaymentMethod	Payment method	Electronic check; Mailed check; Bank transfer (automatic); Credit card (automatic)	feature
MonthlyCharges	Current monthly charge (USD)	non-negative number	feature
TotalCharges	Total billed to date (USD); history only, not added to future value	non-negative number	not a model input
Churn	Left within the observation window; target label, never a feature	Yes; No	target

Derived fields

Field	Type / unit	Definition
churn	0/1	Churn mapped Yes=1, No=0; label only
split	train / validation / test	Role of the row in validation design
Service and account flags	0/1	Partner, Dependents, PhoneService, PaperlessBilling, MultipleLines, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies: 1 if the field is "Yes". "No phone service" and "No internet service" count as 0, so that information is carried only by PhoneService and InternetService

Reference coding	0/1 columns	InternetService against "No"; Contract against "Two year"; PaymentMethod against "Credit card (automatic)" (the lowest-churn level), so each coefficient compares a level with that reference
tenure_band, tenure_segment	new / developing / loyal	tenure <= 12, 13-24, > 24 months; business segment (tested as a model feature, not selected)
churn_probability	probability	Calibrated model output
risk_band	low / medium / high	<= 0.25, 0.25-0.5, > 0.5; descriptive only
prediction_origin	text	in_sample_do_not_evaluate (train), selection_holdout (validation), untouched_test (test), or upload status
remaining_months	months	min(cap 24, base horizon 12 x tenure factor)
discounted_months	months	Present value of remaining_months at 10% a year
customer_value	USD	MonthlyCharges x margin 30% x discounted_months (12-month truncated value proxy, not a validated CLV)
offer_id	text	Offer chosen (or best potential offer if not contacted); see the offer catalog
assumed_save_rate	share	Assumed probability that the offer keeps a would-be churner
offer_cost, contact_cost, total_cost	USD	Cost per contacted customer; total = offer + contact
expected_saved_probability	probability	churn_probability x assumed_save_rate
expected_retained_value	USD	expected_saved_probability x customer_value
expected_net_value	USD	expected_retained_value - total_cost
break_even_save_rate	share	total_cost / (churn_probability x customer_value); save rate at which the offer pays for itself
break_even_probability	probability	total_cost / (save rate x customer_value); churn probability above which the offer pays
selected	bool	Customer is on the contact list
selection_reason, decision_detail	text	Why the customer is in or out, with the EV formula filled in
scenario_id	text	Hash of value settings, offers, budget, strategy, contact cap and customer list
quadrant	text	Risk x value segment using validation 75th-percentile cut-offs

Context fields from the extended sample (never model inputs)

Field	Type / unit	Definition
churn_reason	text, churners only	The customer's stated reason for leaving (20 values), recorded after they left
churn_category	Competitor / Dissatisfaction / Attitude / Price / Other	IBM's grouping of the reasons; Competitor and Price are treated as addressable by a price or contract offer
cltv	index, 7,043 customers	IBM's predicted lifetime-value index

		(undocumented formula); cross-check only

Rules applied to every input file

- Blank TotalCharges with tenure 0 becomes 0; other numeric gaps are filled with training medians.
- Categorical fields must match the allowed values exactly; missing categories are rejected.
- MultipleLines is "No phone service" if and only if PhoneService is "No"; the six internet add-ons are "No internet service" if and only if InternetService is "No".
- A contradiction rejects the whole upload; nothing is partially imported.
- Money is in USD of the sample company (assumed; the source does not state a currency).